Appendix: Detailed Proof of the Modulus Regions for $\mu_\pm$ and Analyticity

This appendix provides a complete derivation of the modulus regions for the two roots $\mu_\pm$ of the quadratic equation associated with the discrete scheme. It also proves the analyticity of the classical and optimized discrete convergence factors ρ^D_{cla} and ρ^D_{opt} in the open right half-plane. These results correspond to the modulus-region and analyticity arguments used in the main text.

We begin by recalling the original expression of the roots $\mu_\pm$. Introducing the variable

$$z = e^{-s\Delta t}, \qquad s = \sigma + i\omega, \qquad \sigma \geq 0,$$

the roots are given by

$$\mu_\pm = \frac{X \pm \sqrt{X^2 - 4a^2|c|^2(\Delta t)^4}}{2a|c|(\Delta t)^2}, \tag{1}$$

where

$$X = 2a|c|(\Delta t)^2 + (1-z)\big(|b|\Delta t + 1 - z\big). \tag{2}$$

Lemma 1 (Modulus regions for $\mu_\pm$)**.** *Let $\mu_\pm$ be defined as in* (1)*, with*

$$z = e^{-s\Delta t}, \qquad s = \sigma + i\omega, \qquad \sigma \geq 0.$$

Introduce

$$C_0 = a|c|(\Delta t)^2, \qquad C_1 = \frac{|b|\Delta t}{C_0}, \qquad C_2 = \frac{1}{C_0},$$

and define

$$\gamma(\sigma) = \frac{1}{4}\left(\frac{C_1}{C_2} + 2\right)e^{\sigma\Delta t},$$

$$r(\sigma) = \sqrt{\frac{1}{2} - \frac{2 + C_1 + C_2}{2C_2}e^{2\sigma\Delta t} + \gamma(\sigma)^2}.$$

Let

$$A = \frac{2 + C_1 + C_2}{2C_2} - \frac{1}{16}\left(\frac{C_1}{C_2} + 2\right)^2.$$

Assume that

$$0 < A \le \frac{1}{2}, \qquad \frac{C_1}{C_2} \le 2,$$

and define

$$\hat{\sigma} = \frac{1}{\Delta t} \log\left(\sqrt{\frac{1}{2A}}\right).$$

Furthermore define

$$\omega_+(\sigma) = \frac{1}{\Delta t} \cos^{-1}\big(\gamma(\sigma) + r(\sigma)\big),$$

$$\omega_-(\sigma) = \frac{1}{\Delta t} \cos^{-1}\big(\gamma(\sigma) - r(\sigma)\big).$$

Then for every $0 \le \sigma \le \hat{\sigma}$ *such that*

$$\gamma(\sigma) \pm r(\sigma) \in [-1, 1],$$

the following assertions hold.

1. *If*

$$\omega \le \omega_+(\sigma) \quad \textit{or} \quad \omega > \omega_-(\sigma),$$

then

$$|\mu_+(z)| > 1, \qquad |\mu_-(z)| < 1.$$

2. *If*

$$\omega_+(\sigma) < \omega \le \omega_-(\sigma),$$

then

$$|\mu_-(z)| > 1, \qquad |\mu_+(z)| < 1,$$

except at $(\sigma, \omega) = (0, 0)$.

3. *Moreover*

$$|\mu_+(z)| = |\mu_-(z)| = 1 \quad \Longleftrightarrow \quad \mu_+(z) = \mu_-(z) = 1 \quad \Longleftrightarrow \quad \sigma = \omega = 0.$$

Proof. We begin by simplifying the expression of X in equation (2) by factoring out the parameter-dependent quantity

$$C_0 = a|c|(\Delta t)^2.$$

We then introduce the normalized coefficients

$$C_1 = \frac{|b|\Delta t}{C_0}, \qquad C_2 = \frac{1}{C_0},$$

so that the quantity X can be written in the normalized form

$$\mathfrak{X}(z) = 2 + C_1(1-z) + C_2(1-z)^2,$$

where

$$z = e^{-s\Delta t}, \qquad s = \sigma + i\omega, \qquad \sigma \geq 0.$$

Accordingly, the roots $\mu_\pm$ take the form

$$\mu_\pm = \frac{\mathfrak{X} \pm \sqrt{\mathfrak{X}^2 - 4}}{2}. \tag{3}$$

To separate real and imaginary parts, write

$$1 - z = \tilde{\sigma} + i\tilde{\omega},$$

where

$$\begin{aligned} \tilde{\sigma} &= 1 - e^{-\sigma\Delta t}\cos(\omega\Delta t), \\ \tilde{\omega} &= e^{-\sigma\Delta t}\sin(\omega\Delta t). \end{aligned} \tag{4}$$

Substituting into $\mathfrak{X}$ gives

$$\mathfrak{X} = x + iy,$$

with

$$\begin{aligned} x &= 2 + C_1\tilde{\sigma} + C_2(\tilde{\sigma}^2 - \tilde{\omega}^2), \\ y &= \tilde{\omega}\,(C_1 + 2C_2\tilde{\sigma}). \end{aligned} \tag{5}$$

From (3), the real parts of μ_+ and μ_- are

$$\begin{aligned} \Re(\mu_+) &= \frac{x}{2} + \frac{1}{4}\sqrt{2\sqrt{x^4 + 2x^2y^2 - 8x^2 + y^4 + 8y^2 + 16} + 2x^2 - 2y^2 - 8}, \\ \Re(\mu_-) &= \frac{x}{2} - \frac{1}{4}\sqrt{2\sqrt{x^4 + 2x^2y^2 - 8x^2 + y^4 + 8y^2 + 16} + 2x^2 - 2y^2 - 8}. \end{aligned} \tag{6}$$

We now distinguish the possible cases according to the signs of x and y.

Case 1: $y \neq 0$ and $x > 0$.
When $x > 0$, the imaginary parts of μ_+ and μ_- are given by

$$\Im(\mu_+) = \frac{y}{2} + \frac{1}{4}\frac{2xy}{|2xy|}\sqrt{2\sqrt{x^4+2x^2y^2-8x^2+y^4+8y^2+16}-2x^2+2y^2+8},$$
$$\Im(\mu_-) = \frac{y}{2} - \frac{1}{4}\frac{2xy}{|2xy|}\sqrt{2\sqrt{x^4+2x^2y^2-8x^2+y^4+8y^2+16}-2x^2+2y^2+8}. \tag{7}$$

Using the definition of x in (5), the condition $x > 0$ is equivalent to

$$2 + C_1\tilde{\sigma} + C_2(\tilde{\sigma}^2 - \tilde{\omega}^2) > 0.$$

Hence

$$\tilde{\omega}^2 < \frac{2 + C_1\tilde{\sigma} + C_2\tilde{\sigma}^2}{C_2}. \tag{8}$$

Substituting

$$\tilde{\sigma} = 1 - e^{-\sigma\Delta t}\cos(\omega\Delta t), \qquad \tilde{\omega} = e^{-\sigma\Delta t}\sin(\omega\Delta t),$$

into (8), we obtain

$$C_2 e^{-2\sigma\Delta t}\sin^2(\omega\Delta t) < 2 + C_1\big(1 - e^{-\sigma\Delta t}\cos(\omega\Delta t)\big) + C_2\big(1 - e^{-\sigma\Delta t}\cos(\omega\Delta t)\big)^2. \tag{9}$$

Rearranging the terms containing $\cos(\omega\Delta t)$ gives

$$C_2 e^{-2\sigma\Delta t} - (2 + C_1 + C_2) < 2C_2 e^{-2\sigma\Delta t}\cos^2(\omega\Delta t) - (C_1 + 2C_2)e^{-\sigma\Delta t}\cos(\omega\Delta t). \tag{10}$$

Dividing by $2C_2 e^{-2\sigma\Delta t} > 0$, we get

$$\frac{1}{2} - \frac{2 + C_1 + C_2}{2C_2}e^{2\sigma\Delta t} < \cos^2(\omega\Delta t) - \frac{C_1 + 2C_2}{2C_2}e^{\sigma\Delta t}\cos(\omega\Delta t). \tag{11}$$

Completing the square on the right-hand side yields

$$\begin{aligned}\frac{1}{2} - \frac{2 + C_1 + C_2}{2C_2}e^{2\sigma\Delta t} < \left(\cos(\omega\Delta t) - \frac{1}{4}\frac{C_1 + 2C_2}{C_2}e^{\sigma\Delta t}\right)^2 \\ - \frac{1}{16}\left(\frac{C_1 + 2C_2}{C_2}\right)^2 e^{2\sigma\Delta t}.\end{aligned} \tag{12}$$

Hence

$$\left|\cos(\omega\Delta t) - \frac{1}{4}\frac{C_1 + 2C_2}{C_2}e^{\sigma\Delta t}\right| > \sqrt{\frac{1}{2} - \frac{2 + C_1 + C_2}{2C_2}e^{2\sigma\Delta t} + \frac{1}{16}\left(\frac{C_1 + 2C_2}{C_2}\right)^2 e^{2\sigma\Delta t}}. \tag{13}$$

Define

$$\gamma(\sigma) = \frac{1}{4}\frac{C_1 + 2C_2}{C_2} e^{\sigma \Delta t},$$

and

$$A = \frac{2 + C_1 + C_2}{2C_2} - \frac{1}{16}\left(\frac{C_1}{C_2} + 2\right)^2.$$

Then the square root in (13) becomes

$$r(\sigma) = \sqrt{\frac{1}{2} - Ae^{2\sigma\Delta t}}.$$

Therefore (13) becomes

$$|\cos(\omega\Delta t) - \gamma(\sigma)| > r(\sigma),$$

that is,

$$\cos(\omega\Delta t) > \gamma(\sigma) + r(\sigma) \quad \text{or} \quad \cos(\omega\Delta t) < \gamma(\sigma) - r(\sigma).$$

At this stage, the admissibility conditions become essential, since the boundary curves are defined through the expressions

$$\omega_\pm(\sigma) = \frac{1}{\Delta t}\cos^{-1}\big(\gamma(\sigma) \pm r(\sigma)\big),$$

with

$$r(\sigma) = \sqrt{\frac{1}{2} - Ae^{2\sigma\Delta t}}.$$

For these boundary curves to be real-valued, two conditions must hold simultaneously

1. the square root defining $r(\sigma)$ must be real;
2. the arguments of the inverse cosine must belong to the interval $[-1, 1]$.

We first analyze the square root condition. Since $r(\sigma)$ is real only when its radicand is nonnegative, we require

$$\frac{1}{2} - Ae^{2\sigma\Delta t} \geq 0.$$

If $A > 0$, this inequality is equivalent to

$$Ae^{2\sigma\Delta t} \leq \frac{1}{2}.$$

Because $e^{2\sigma\Delta t} \geq 1$ for every $\sigma \geq 0$, the above inequality can hold on a nonempty interval only if

$$A \leq \frac{1}{2}.$$

Thus, in the positive-A case, the admissible regime is

$$0 < A \leq \frac{1}{2}.$$

Solving

$$\frac{1}{2} - Ae^{2\sigma\Delta t} = 0$$

for σ gives the threshold at which the radicand vanishes

$$Ae^{2\sigma\Delta t} = \frac{1}{2} \quad \Longleftrightarrow \quad e^{2\sigma\Delta t} = \frac{1}{2A} \quad \Longleftrightarrow \quad 2\sigma\Delta t = \log\left(\frac{1}{2A}\right).$$

Hence

$$\sigma = \frac{1}{2\Delta t}\log\left(\frac{1}{2A}\right) = \frac{1}{\Delta t}\log\left(\sqrt{\frac{1}{2A}}\right).$$

We therefore define

$$\hat{\sigma} = \frac{1}{\Delta t}\log\left(\sqrt{\frac{1}{2A}}\right),$$

and obtain the admissible interval

$$0 \leq \sigma \leq \hat{\sigma}.$$

Next we examine the inverse-cosine condition. In order that $\omega_{\pm}(\sigma)$ be real, we must also have

$$\gamma(\sigma) + r(\sigma) \in [-1, 1], \qquad \gamma(\sigma) - r(\sigma) \in [-1, 1].$$

From the separate reality analysis of the boundary curves, a necessary condition for this is

$$\frac{C_1}{C_2} \leq 2.$$

Indeed, writing

$$\gamma(\sigma) = \gamma_0 e^{\sigma\Delta t}, \qquad \gamma_0 = \frac{1}{4}\left(\frac{C_1}{C_2} + 2\right),$$

we see that $\gamma(\sigma)$ is increasing in σ, so its smallest value occurs at $\sigma = 0$. Therefore, if $\gamma(0) > 1$, then $\gamma(\sigma) > 1$ for all $\sigma \geq 0$, and the inverse cosine cannot be real. The condition

$$\gamma(0) = \frac{1}{4}\left(\frac{C_1}{C_2} + 2\right) \leq 1$$

is therefore necessary, and this is equivalent to

$$\frac{C_1}{C_2} \leq 2.$$

Finally, we analyze the case $A \leq 0$. Recall that the constant A is defined by

$$A = \frac{2 + C_1 + C_2}{2C_2} - \frac{1}{16}\left(\frac{C_1}{C_2} + 2\right)^2.$$

To understand the consequences of the condition $A \leq 0$, we rewrite this expression in terms of the ratio

$$t = \frac{C_1}{C_2}.$$

Substituting $C_1 = tC_2$ into the expression for A gives

$$A = \frac{2 + tC_2 + C_2}{2C_2} - \frac{1}{16}(t+2)^2 = \frac{2 + (t+1)C_2}{2C_2} - \frac{1}{16}(t+2)^2.$$

The inequality $A \leq 0$ is therefore equivalent to

$$\frac{2 + (t+1)C_2}{2C_2} \leq \frac{1}{16}(t+2)^2.$$

Multiplying both sides by $16C_2 > 0$ yields

$$8\big(2 + (t+1)C_2\big) \leq C_2(t+2)^2.$$

Expanding both sides gives

$$16 + 8(t+1)C_2 \leq C_2(t^2 + 4t + 4).$$

Rearranging terms leads to

$$16 + 8(t+1)C_2 - C_2(t^2 + 4t + 4) \leq 0,$$

or equivalently

$$16 + 4C_2(t+1) - C_2t^2 \leq 0.$$

Since $C_2 > 0$, this inequality implies

$$C_2t^2 - 4C_2(t+1) - 16 \geq 0.$$

Dividing by C_2 gives

$$t^2 - 4t - 4 - \frac{16}{C_2} \geq 0.$$

The left-hand side is a quadratic polynomial in t. Because $C_2 > 0$, the constant term $-4 - \frac{16}{C_2}$ is strictly negative, and therefore the inequality can hold only when t is sufficiently large. In particular, a necessary condition is

$$t > 2.$$

Recalling that $t = \frac{C_1}{C_2}$, we conclude that

$$A \leq 0 \quad \Longrightarrow \quad \frac{C_1}{C_2} > 2.$$

This relation has an important consequence for the boundary curves. Indeed, from the definition

$$\gamma(\sigma) = \frac{1}{4}\left(\frac{C_1}{C_2} + 2\right) e^{\sigma \Delta t},$$

we obtain

$$\gamma(0) = \frac{1}{4}\left(\frac{C_1}{C_2} + 2\right).$$

If $\frac{C_1}{C_2} > 2$, then

$$\gamma(0) > 1.$$

Since $e^{\sigma \Delta t} \geq 1$ for all $\sigma \geq 0$, it follows that

$$\gamma(\sigma) = \gamma(0)e^{\sigma \Delta t} \geq \gamma(0) > 1.$$

Consequently,

$$\gamma(\sigma) + r(\sigma) > 1 \qquad \text{for all } \sigma \geq 0.$$

Therefore the arguments of the inverse cosine lie outside the interval $[-1, 1]$, so the expressions

$$\cos^{-1}\big(\gamma(\sigma) \pm r(\sigma)\big)$$

are not real-valued. In this regime the boundary curves cannot be defined in the real frequency domain.

For this reason the parameter regime $A \le 0$ must be excluded from the admissible region of the analysis.
Consequently, the boundary curves are real-valued only in the regime

$$0 < A \le \frac{1}{2}, \qquad \frac{C_1}{C_2} \le 2, \qquad 0 \le \sigma \le \hat{\sigma},$$

together with

$$\gamma(\sigma) + r(\sigma) \in [-1, 1], \qquad \gamma(\sigma) - r(\sigma) \in [-1, 1].$$

Since

$$z = e^{-s\Delta t} = e^{-\sigma\Delta t} e^{-i\omega\Delta t},$$

all expressions depend on the phase $\omega\Delta t$. It is therefore sufficient to work on the principal interval

$$0 \le \omega\Delta t \le \pi, \qquad \text{i.e.} \qquad 0 \le \omega \le \frac{\pi}{\Delta t}.$$

On this interval, the cosine function is strictly decreasing and hence invertible. Applying $\cos^{-1}$ gives

$$\omega < \frac{1}{\Delta t} \cos^{-1}\big(\gamma(\sigma) + r(\sigma)\big) \quad \text{or} \quad \omega > \frac{1}{\Delta t} \cos^{-1}\big(\gamma(\sigma) - r(\sigma)\big).$$

We now compare the moduli of μ_+ and μ_-. From (6) and (7), when $x > 0$ and $y \neq 0$, we have

$$\big(\Re(\mu_+)\big)^2 > \big(\Re(\mu_-)\big)^2, \qquad \big(\Im(\mu_+)\big)^2 > \big(\Im(\mu_-)\big)^2.$$

Therefore

$$|\mu_+|^2 = \big(\Re(\mu_+)\big)^2 + \big(\Im(\mu_+)\big)^2 > \big(\Re(\mu_-)\big)^2 + \big(\Im(\mu_-)\big)^2 = |\mu_-|^2.$$

By Vieta's formula,

$$\mu_+\mu_- = 1, \qquad \text{hence} \qquad |\mu_+|\,|\mu_-| = 1.$$

It follows that

$$|\mu_+| > 1, \qquad |\mu_-| < 1.$$

Case 2: $y \neq 0$ and $x < 0$.

When $x < 0$, the inequalities derived above yield

$$\gamma(\sigma) - r(\sigma) < \cos(\omega\Delta t) < \gamma(\sigma) + r(\sigma),$$

and therefore, after applying $\cos^{-1}$,

$$\frac{1}{\Delta t}\cos^{-1}\!\big(\gamma(\sigma) + r(\sigma)\big) < \omega < \frac{1}{\Delta t}\cos^{-1}\!\big(\gamma(\sigma) - r(\sigma)\big).$$

For $x < 0$, the real and imaginary parts satisfy

$$\big(\Re(\mu_+)\big)^2 = \left(\frac{1}{2}|x| - \frac{1}{4}\sqrt{2\sqrt{x^4 + 2x^2y^2 - 8x^2 + y^4 + 8y^2 + 16} + 2x^2 - 2y^2 - 8}\right)^2,$$
$$\big(\Re(\mu_-)\big)^2 = \left(\frac{1}{2}|x| + \frac{1}{4}\sqrt{2\sqrt{x^4 + 2x^2y^2 - 8x^2 + y^4 + 8y^2 + 16} + 2x^2 - 2y^2 - 8}\right)^2,$$

and

$$\big(\Im(\mu_+)\big)^2 = \left(\frac{1}{2}|y| - \frac{1}{4}\sqrt{2\sqrt{x^4 + 2x^2y^2 - 8x^2 + y^4 + 8y^2 + 16} - 2x^2 + 2y^2 + 8}\right)^2,$$
$$\big(\Im(\mu_-)\big)^2 = \left(\frac{1}{2}|y| + \frac{1}{4}\sqrt{2\sqrt{x^4 + 2x^2y^2 - 8x^2 + y^4 + 8y^2 + 16} - 2x^2 + 2y^2 + 8}\right)^2.$$

Hence

$$\big(\Re(\mu_+)\big)^2 < \big(\Re(\mu_-)\big)^2, \qquad \big(\Im(\mu_+)\big)^2 < \big(\Im(\mu_-)\big)^2,$$

so

$$|\mu_+|^2 < |\mu_-|^2.$$

Using again

$$|\mu_+|\,|\mu_-| = 1,$$

we conclude that

$$|\mu_+| < 1, \qquad |\mu_-| > 1.$$

Case 3: $x = 0$**.**
Suppose that $x = 0$. From (5) this condition corresponds exactly to the boundary curves separating the two modulus regions, which are given by

$$\omega = \frac{1}{\Delta t}\cos^{-1}\!\big(\gamma(\sigma) \pm r(\sigma)\big).$$

In this situation we have

$$\mathfrak{X} = iy,$$

and the roots $\mu_\pm$ defined in (3) become

$$\mu_\pm = \frac{iy \pm \sqrt{-y^2-4}}{2} = \frac{i}{2}\big(y \pm \sqrt{y^2+4}\big),$$

where the principal branch of the square root is used.

Therefore

$$|\mu_+|^2 = \frac{y^2}{2} + \frac{y}{2}\sqrt{y^2+4} + 1, \qquad |\mu_-|^2 = \frac{y^2}{2} - \frac{y}{2}\sqrt{y^2+4} + 1.$$

We now examine the sign of y. From (5) we have

$$y = \tilde{\omega}\big(C_1 + 2C_2\tilde{\sigma}\big).$$

Since $C_1 > 0$, $C_2 > 0$, and $0 \leq \tilde{\sigma} \leq 2$, the factor $C_1 + 2C_2\tilde{\sigma}$ is strictly positive.

Moreover, recalling the definition of $\tilde{\omega}$ in (4), and noting that $0 \leq \omega\Delta t \leq \pi$, we have

$$\sin(\omega\Delta t) \geq 0,$$

which implies

$$\tilde{\omega} \geq 0.$$

Consequently

$$y = \tilde{\omega}(C_1 + 2C_2\tilde{\sigma}) \geq 0,$$

so the case $y < 0$ does not occur in the admissible frequency interval.

The equality $y = 0$ requires $\tilde{\omega} = 0$, which occurs only when $\omega = 0$. Together with the condition $x = 0$, this gives the unique point

$$(\sigma, \omega) = (0, 0),$$

which will be treated separately below. Hence, along the boundary $x = 0$ we have $y > 0$.

Since $y > 0$, the expressions above give

$$|\mu_+| > |\mu_-|.$$

Using Vieta's relation for the quadratic equation,

$$\mu_+\mu_- = 1,$$

we obtain

$$|\mu_+|\,|\mu_-| = 1.$$

Hence

$$|\mu_+| > 1, \qquad |\mu_-| < 1.$$

Thus, on the boundary $x = 0$ we have

$$|\mu_+| > 1, \qquad |\mu_-| < 1,$$

except at the single point $(\sigma, \omega) = (0, 0)$ where

$$\mu_+ = \mu_- = 1.$$

Case 4: $y = 0$ **with** $x \neq 0$**.**
In this case we assume $y = 0$. From (5) we have

$$y = \tilde{\omega}(C_1 + 2C_2\tilde{\sigma}),$$

and since $C_1 > 0$ and $C_2 > 0$, this occurs if and only if

$$\tilde{\omega} = 0.$$

Hence $x \neq 0$ and the roots $\mu_\pm$ become

$$\mu_\pm = \frac{2 + C_1\tilde{\sigma} + C_2\tilde{\sigma}^2 \pm \sqrt{(2 + C_1\tilde{\sigma} + C_2\tilde{\sigma}^2)^2 - 4}}{2}.$$

If $\tilde{\sigma} = 0$, then

$$1 - e^{-\sigma\Delta t}\cos(\omega\Delta t) = 0,$$

which implies

$$\cos(\omega\Delta t) = e^{\sigma\Delta t}.$$

Since $|\cos(\cdot)| \leq 1$ and $e^{\sigma\Delta t} \geq 1$ for $\sigma \geq 0$, this occurs only when

$$\sigma = 0, \qquad \omega = 0.$$

Hence

$$\mu_+ = \mu_- = 1.$$

The only other case where $\mu_+ = \mu_- = 1$ would be when

$$\tilde{\sigma} = -\frac{C_1}{C_2},$$

which is not admissible since $0 \leq \tilde{\sigma} \leq 2$.

When $0 < \tilde{\sigma} \leq 2$ (i.e., $\sigma > 0$), we obtain

$$\mu_+ > \mu_-,$$

which implies

$$|\mu_+| > |\mu_-|,$$

and therefore

$$|\mu_+| > 1, \qquad |\mu_-| < 1.$$

Final step. It remains to identify the unique point, if any, at which both roots have unit modulus. For the final part of the proof we determine when the roots $\mu_\pm$ defined in (3) satisfy $\mu_\pm = 1$. We first examine the root μ_-. From (3) we have

$$\mu_- = \frac{\mathfrak{X} - \sqrt{\mathfrak{X}^2 - 4}}{2}.$$

Imposing $\mu_- = 1$ gives

$$\frac{\mathfrak{X} - \sqrt{\mathfrak{X}^2 - 4}}{2} = 1 \quad \Longleftrightarrow \quad \mathfrak{X} - 2 = \sqrt{\mathfrak{X}^2 - 4}.$$

Squaring both sides yields

$$(\mathfrak{X} - 2)^2 = \mathfrak{X}^2 - 4 \quad \Longleftrightarrow \quad \mathfrak{X} = 2.$$

Using the definition of $\mathfrak{X}$,

$$\mathfrak{X} = 2 + C_1(1 - z) + C_2(1 - z)^2,$$

we see that $\mathfrak{X} = 2$ if and only if

$$1 - z = 0 \qquad \text{or} \qquad 1 - z = -\frac{C_1}{C_2}.$$

At such a point $\mathfrak{X}$ must be real, and therefore $y = 0$, equivalently $\tilde{\omega} = 0$. Therefore the roots are real, and squaring the equation introduces no extraneous solutions.

The second possibility would give

$$\tilde{\sigma} = -\frac{C_1}{C_2},$$

which is not admissible since $0 \le \tilde{\sigma} \le 2$. Hence the only admissible solution is

$$1 - z = 0, \qquad \text{i.e.,} \qquad z = 1.$$

Since $z = e^{-s\Delta t}$, this occurs only when

$$s = 0, \qquad \text{that is} \qquad \sigma = 0, \quad \omega = 0.$$

A similar argument applies to μ_+. Therefore the equation

$$\mu_\pm = 1$$

is satisfied only at the single point

$$(\sigma, \omega) = (0, 0),$$

where

$$\mu_+ = \mu_- = 1.$$

Collecting the above cases, we conclude that in the admissible regime

$$0 < A \le \frac{1}{2}, \qquad \frac{C_1}{C_2} \le 2, \qquad 0 \le \sigma \le \hat{\sigma},$$

with

$$\gamma(\sigma) \pm r(\sigma) \in [-1, 1],$$

the modulus of the roots is partitioned as follows.

If

$$\omega \le \frac{1}{\Delta t} \cos^{-1}\big(\gamma(\sigma) + r(\sigma)\big) \quad \text{or} \quad \omega > \frac{1}{\Delta t} \cos^{-1}\big(\gamma(\sigma) - r(\sigma)\big),$$

then

$$|\mu_+(z)| > 1, \qquad |\mu_-(z)| < 1.$$

If instead

$$\frac{1}{\Delta t} \cos^{-1}\big(\gamma(\sigma) + r(\sigma)\big) < \omega < \frac{1}{\Delta t} \cos^{-1}\big(\gamma(\sigma) - r(\sigma)\big),$$

then

$$|\mu_-(z)| > 1, \qquad |\mu_+(z)| < 1.$$

Finally,

$$|\mu_+(z)| = |\mu_-(z)| = 1 \quad \Longleftrightarrow \quad \mu_+(z) = \mu_-(z) = 1 \quad \Longleftrightarrow \quad \sigma = \omega = 0.$$

This completes the proof. $\square$

We now introduce the classical and optimized discrete convergence factors used in the analyticity analysis.

$$\rho_{\text{cla}}^D(s) = \begin{cases} \mu_-^2, & |\mu_+(s)| > 1, \\ \mu_+^2, & |\mu_+(s)| < 1, \end{cases} \tag{14}$$

$$\rho_{\text{opt}}^D(s;\alpha) = \begin{cases} \left(\frac{\mu_-(1-z)+\alpha\Delta t\, c\,(\mu_- -1)}{(1-z)+\alpha\Delta t\, c\,(1-\mu_-)}\right)^2, & |\mu_+(s)| > 1, \\ \left(\frac{\mu_+(1-z)+\alpha\Delta t\, c\,(\mu_+ -1)}{(1-z)+\alpha\Delta t\, c\,(1-\mu_+)}\right)^2, & |\mu_+(s)| < 1, \end{cases} \tag{15}$$

where $z = e^{-s\Delta t}$ and

$$\mu_\pm = \frac{X \pm \sqrt{X^2 - 4a^2|c|^2(\Delta t)^4}}{2a|c|(\Delta t)^2}, \tag{16}$$

with

$$X = 2a|c|(\Delta t)^2 + (1-z)(|b|\Delta t + 1 - z). \tag{17}$$

Analyticity of the Discrete Convergence Factors

We now prove that the classical and optimized discrete convergence factors admit single analytic representations in the open right half-plane.

Theorem 1 (Analytic square root)**.** *Let $D \subset \mathbb{C}$ be a simply connected domain, and let f be analytic and nonvanishing in D. Then there exists an analytic function g in D such that*

$$g(z)^2 = f(z), \qquad z \in D.$$

Proof. See Conway [1, Theorem 2.2(h), p. 202]. $\square$

Lemma 2 (Analyticity of $\rho_{\text{cla}}^D(s)$ in the right half-plane)**.** *Assume that*

$$a > 0, \qquad b < 0, \qquad c < 0, \qquad \Delta t > 0.$$

Let the classical discrete convergence factor $\rho_{\text{cla}}^D(s)$ be defined by (14), where the characteristic roots $\mu_\pm$ are given by (16)–(17).
Then the classical discrete convergence factor $\rho_{\text{cla}}^D(s)$ is analytic in the open right half-plane $\Re(s) > 0$.

Proof. Let
$$H = \{\, s \in \mathbb{C} : \Re(s) > 0 \,\}, \qquad z = e^{-s\Delta t}.$$
Then
$$|z| = e^{-\Re(s)\Delta t} < 1,$$
so that
$$z \in D := \{\, z \in \mathbb{C} : |z| < 1 \,\}.$$
Set
$$w = 1 - z.$$
Then
$$\Re(w) = 1 - \Re(z) \geq 1 - |z| > 0.$$
Using
$$X = 2a|c|(\Delta t)^2 + (1-z)(|b|\Delta t + 1 - z),$$
we define
$$\mathfrak{X}(z) = \frac{X}{a|c|(\Delta t)^2}.$$
Hence
$$\mathfrak{X}(z) = 2 + C_1(1-z) + C_2(1-z)^2,$$
where
$$C_1 = \frac{|b|\Delta t}{a|c|(\Delta t)^2}, \qquad C_2 = \frac{1}{a|c|(\Delta t)^2}.$$
Since $a > 0$, $c < 0$, and $\Delta t > 0$, we have
$$C_1 > 0, \qquad C_2 > 0.$$
Moreover, since $z = e^{-s\Delta t}$ is analytic, $\mathfrak{X}(z)$ is analytic in H.

We next show that
$$\mathfrak{X}(z)^2 - 4 \neq 0 \qquad \text{for all } s \in H.$$
Write
$$w = u + iv, \qquad u = \Re(w) > 0.$$
Then
$$\mathfrak{X}(z) = 2 + C_1(u+iv) + C_2(u+iv)^2.$$
Therefore,
$$\Im(\mathfrak{X}(z)) = v(C_1 + 2C_2 u).$$

Since
$$C_1 > 0, \qquad C_2 > 0, \qquad u > 0,$$
we obtain
$$C_1 + 2C_2u > 0.$$
Hence, if $\mathfrak{X}(z)$ is real, then necessarily $v = 0$. Thus
$$w = u > 0,$$
and therefore
$$\mathfrak{X}(z) = 2 + C_1u + C_2u^2 > 2.$$
Consequently,
$$\mathfrak{X}(z) \neq 2, \qquad \mathfrak{X}(z) \neq -2,$$
and hence
$$\mathfrak{X}(z)^2 - 4 \neq 0.$$

Since the unit disk
$$D = \{\, z \in \mathbb{C} : |z| < 1 \,\}$$
is simply connected and
$$\mathfrak{X}(z)^2 - 4$$
is analytic and nonvanishing in D, Theorem 1 implies that there exists an analytic branch
$$Q(z) = \sqrt{\mathfrak{X}(z)^2 - 4}$$
in D.
Define
$$\mu_0(z) = \frac{\mathfrak{X}(z) - Q(z)}{2}, \qquad \mu_1(z) = \frac{\mathfrak{X}(z) + Q(z)}{2}.$$
Then μ_0 and μ_1 are analytic in D, and
$$\mu_0(z)\mu_1(z) = 1.$$

Choose the branch $Q(z)$ so that for real $s = \sigma > 0$,
$$Q(z) > 0.$$

For real $\sigma > 0$,
$$z = e^{-\sigma \Delta t} \in (0, 1),$$
and hence
$$w = 1 - z > 0.$$

Therefore,

$$\mathfrak{X}(z) = 2 + C_1 w + C_2 w^2 > 2.$$

Thus

$$\mu_0(z) = \frac{\mathfrak{X}(z) - \sqrt{\mathfrak{X}(z)^2 - 4}}{2}$$

satisfies

$$0 < \mu_0(z) < 1.$$

We now show that

$$|\mu_0(z)| < 1 \qquad \text{for all } z \in D.$$

Suppose, for contradiction, that there exists $z_1 \in D$ such that

$$|\mu_0(z_1)| \geq 1.$$

Since D is connected and μ_0 is continuous, there exists $z_* \in D$ such that

$$|\mu_0(z_*)| = 1.$$

Using

$$\mu_0(z_*)\mu_1(z_*) = 1,$$

we also obtain

$$|\mu_1(z_*)| = 1.$$

Thus, there exists $\theta \in \mathbb{R}$ such that

$$\mu_0(z_*) = e^{i\theta}, \qquad \mu_1(z_*) = e^{-i\theta}.$$

Therefore,

$$\mathfrak{X}(z_*) = \mu_0(z_*) + \mu_1(z_*) = 2\cos\theta \in [-2, 2].$$

Hence $\mathfrak{X}(z_*)$ is real. However, we already proved that whenever $\mathfrak{X}(z)$ is real,

$$\mathfrak{X}(z) > 2.$$

This contradiction shows that

$$|\mu_0(z)| < 1 \qquad \text{for all } z \in D.$$

Since

$$\mu_0(z)\mu_1(z) = 1,$$

it follows that

$$|\mu_1(z)| > 1 \qquad \text{for all } z \in D.$$

Hence $\mu_0(z)$ is the unique stable root.

The piecewise definition of $\rho^D_{\text{cla}}(s)$ selects precisely this stable branch. Indeed,

$$\mu_0(z) = \mu_-(z) \quad \text{when } |\mu_+(z)| > 1,$$

and

$$\mu_0(z) = \mu_+(z) \quad \text{when } |\mu_+(z)| < 1.$$

Therefore,

$$\rho^D_{\text{cla}}(s) = \mu_0(z)^2.$$

Since $\mu_0(z)$ is analytic in D and $z = e^{-s\Delta t}$ is analytic in H, it follows that $\rho^D_{\text{cla}}(s)$ is analytic in the open right half-plane $\Re(s) > 0$.

$\square$

Lemma 3 (Analyticity of $\rho^D_{\text{opt}}(s)$ in the right half-plane)**.** *Assume that*

$$a > 0, \qquad b < 0, \qquad c < 0, \qquad \Delta t > 0,$$

and let

$$\alpha < 0.$$

Let the optimized discrete convergence factor $\rho^D_{\text{opt}}(s)$ be defined by (15)*, where the characteristic roots $\mu_\pm$ are given by* (16)*–*(17)*.*
Then the optimized discrete convergence factor $\rho^D_{\text{opt}}(s)$ is analytic in the open right half-plane $\Re(s) > 0$.

Proof. Let

$$H = \{\, s \in \mathbb{C} : \Re(s) > 0 \,\}, \qquad z = e^{-s\Delta t}.$$

By Lemma 2, there exists a unique analytic stable branch

$$\mu_0(z)$$

such that

$$|\mu_0(z)| < 1 \qquad \text{for all } z \in D.$$

Moreover,

$$\mu_0(z) = \mu_-(z) \quad \text{when } |\mu_+(z)| > 1,$$

and

$$\mu_0(z) = \mu_+(z) \quad \text{when } |\mu_+(z)| < 1.$$

Hence, the piecewise definition of $\rho^D_{\text{opt}}(s)$ can be written globally as

$$\rho^D_{\text{opt}}(s) = \left(\frac{\mu_0(1-z) + \alpha \Delta t\, c\, (\mu_0 - 1)}{(1-z) + \alpha \Delta t\, c\, (1 - \mu_0)} \right)^2.$$

Since $z = e^{-s\Delta t}$ and $\mu_0(z)$ are analytic, it remains only to show that the denominator does not vanish in H.
Assume, for contradiction, that

$$(1 - z) + \alpha \Delta t\, c\, (1 - \mu_0) = 0.$$

Then

$$1 - \mu_0 = \frac{1 - z}{-\alpha \Delta t\, c}.$$

Since $\alpha < 0, \qquad c < 0$, we have

$$-\alpha \Delta t\, c < 0.$$

Because $\Re(1 - z) > 0$, it follows that

$$\Re(1 - \mu_0) < 0.$$

Hence

$$\Re(\mu_0) > 1,$$

which contradicts

$$|\mu_0| < 1.$$

Therefore,

$$(1 - z) + \alpha \Delta t\, c\, (1 - \mu_0) \neq 0.$$

Consequently, $\rho^D_{\text{opt}}(s)$ is analytic in the open right half-plane $\Re(s) > 0$.

$\square$

References

[1] J. B. Conway, *Functions of One Complex Variable I*, 2nd ed., Graduate Texts in Mathematics, vol. 11, Springer, New York, 1978.